

- 1 Hydrogencarbonate indicator is a water-soluble solution that can act as a source of carbon dioxide for aquatic photosynthetic organisms. The solution changes colour depending on the concentration of carbon dioxide in the solution. These colours are related to different pH values, as shown in Table 1.1.

Table 1.1

colour of hydrogencarbonate indicator solution	pH	concentration of carbon dioxide in the solution
yellow	7.6	↑ increasing carbon dioxide concentration
yellow-orange	7.8	
orange	8.0	
orange-red	8.2	
red	8.4	atmospheric concentration
red-magenta	8.6	↓ decreasing carbon dioxide concentration
magenta	8.8	
magenta-purple	9.0	
purple	9.2	

Chlorella vulgaris is a protocist that is single-celled, aquatic and photosynthetic. It can be immobilised in alginate beads.

Alginate beads with immobilised *C. vulgaris* can be used to measure the rate of photosynthesis.

- (a) A student noticed that a colour change occurred, from red to magenta, when the alginate beads with immobilised *C. vulgaris* were left in a container of hydrogencarbonate indicator solution and exposed to light.

Explain why this colour change occurred.

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- (b) The student used the alginate beads with immobilised *C. vulgaris* in hydrogencarbonate indicator solution to investigate the rate of photosynthesis in different light intensities.

Fig. 1.1 shows some of the apparatus and reagents the student used.

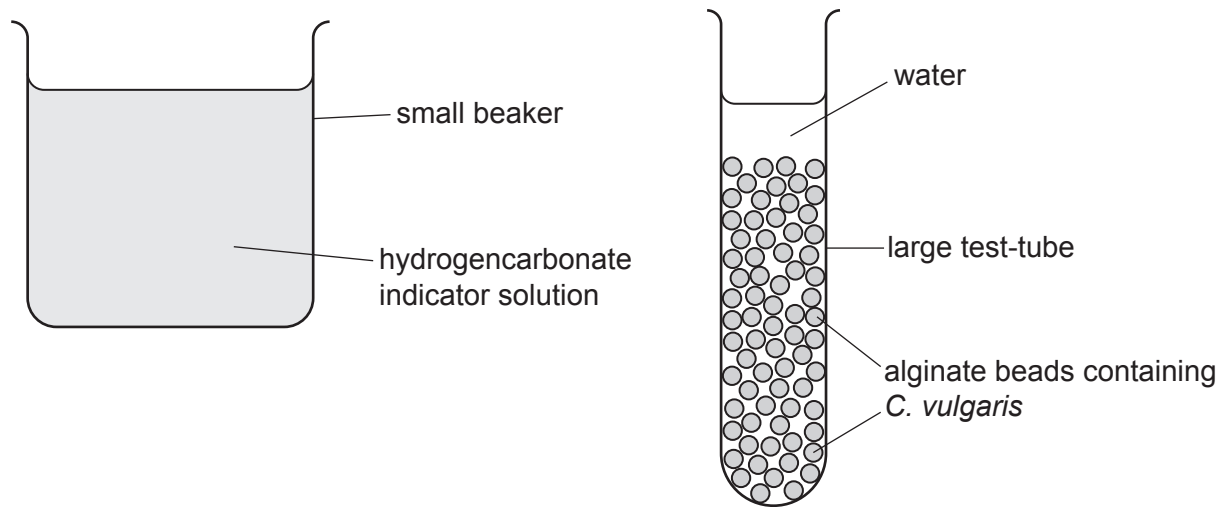


Fig. 1.1

- (i) Identify the **independent** variable in this investigation.

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- (ii) The student was provided with a supply of alginate beads containing immobilised *C. vulgaris*.

Describe a method the student could use to collect data to determine the effect of light intensity on the rate of photosynthesis of *C. vulgaris* using hydrogencarbonate indicator and the experimental set-up in Fig. 1.1.

Your method should be set out in a logical order and be detailed enough to let another person follow.

[8]

- (c) The student set up a large test-tube containing alginate beads with immobilised *C. vulgaris* in hydrogencarbonate indicator solution at **pH 8.4** (red).

The student kept this set-up in the dark for 12 hours.

Predict and explain the results that will be observed after 12 hours in the dark.

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- (d) Some scientists wanted to culture cells of *C. vulgaris* on a large scale for use as a biofuel.

To determine the optimal growing conditions for *C. vulgaris*, the scientists needed to determine the number of cells per cm^3 of suspension to monitor the population growth.

They tried two methods to determine the number of cells per cm^3 of suspension.

The first method used a Secchi stick, as shown in Fig. 1.2.

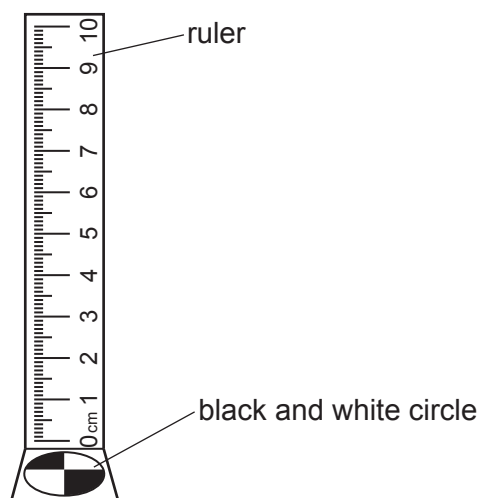


Fig. 1.2

The Secchi stick is lowered into the suspension of cells until the black and white circle is not able to be seen from above.

The depth in cm is recorded from the ruler, as shown in Fig 1.3.

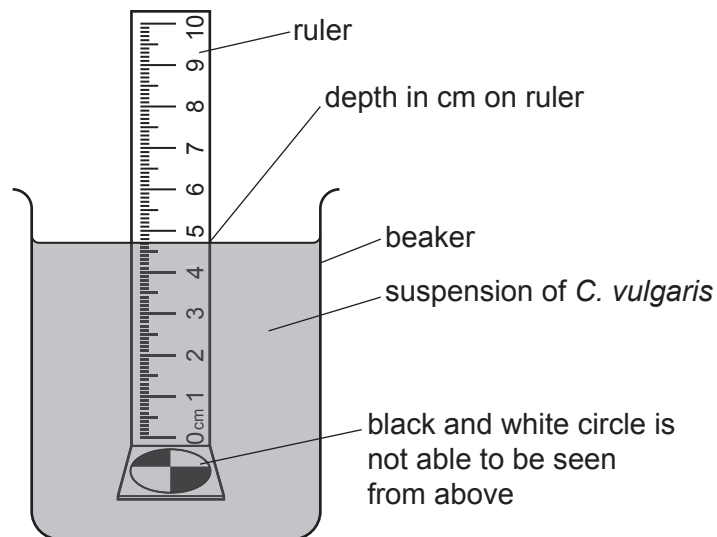


Fig. 1.3

The $\log_{10}$ (lg) of the number of cells is determined from a graph of $\log_{10}$ of cells counted per cm^3 suspension against Secchi depth (cm), as shown in Fig 1.4.

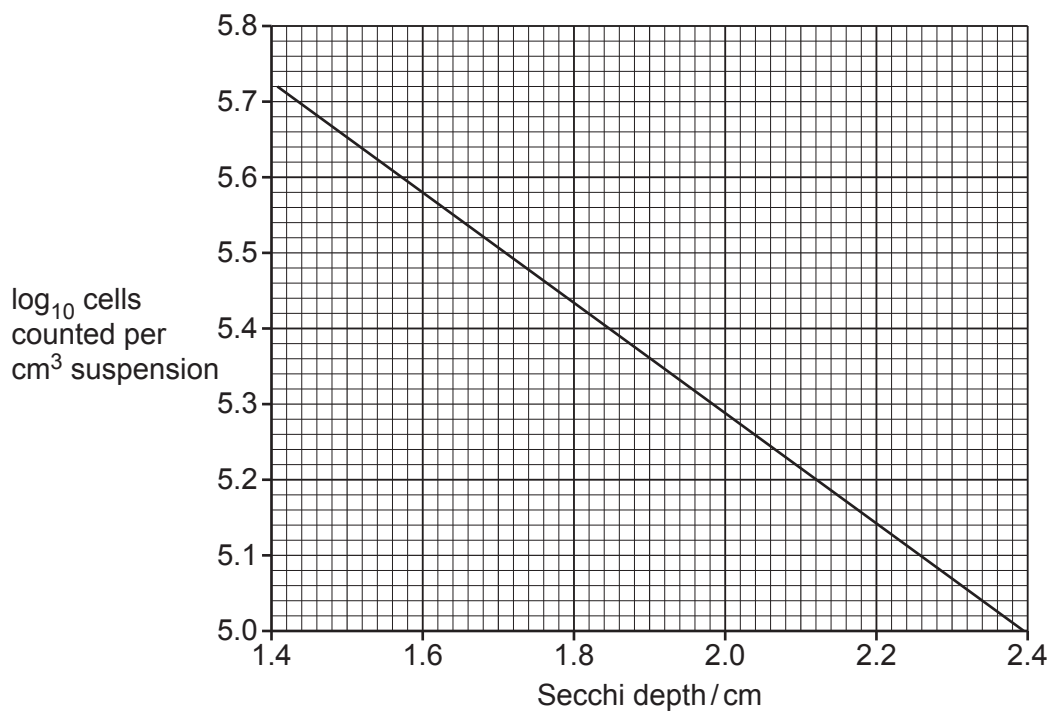


Fig. 1.4

- (i) When the scientists inserted the Secchi stick into a sample from their cell suspension, the circle (on the Secchi stick) was not able to be seen at a depth of **1.9 cm**.

Using the graph in Fig. 1.4, calculate the actual number of cells per cm^3 of suspension.

Show your working and give your answer to the nearest 1000 cells.

number of cells per cm^3 of suspension [2]

The second method used a counting chamber to determine the number of cells per cm^3 of suspension.

Fig. 1.5 shows a section of a counting chamber with cells present, as viewed using the high power of a light microscope.

The depth of the $1\text{ mm} \times 1\text{ mm}$ counting chamber is **0.1 mm**.

The scientists counted the number of cells in several sections of a counting chamber.

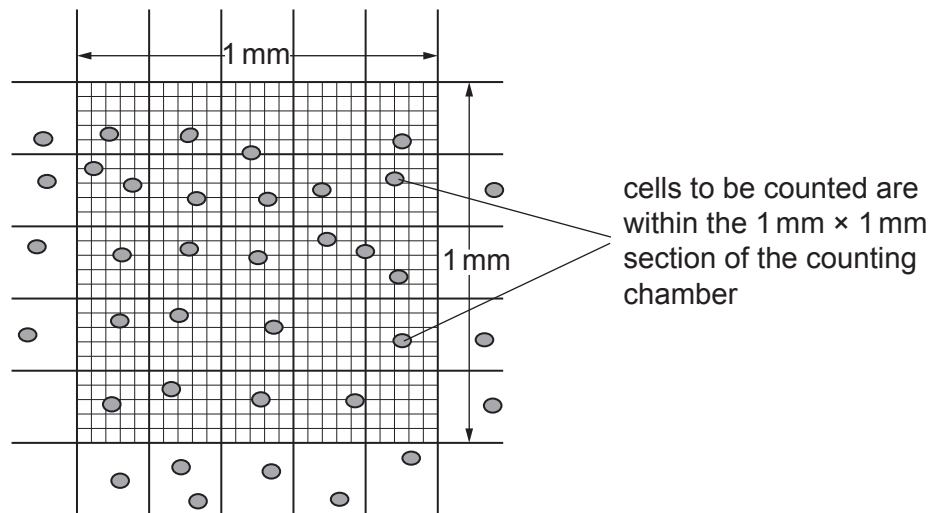


Fig. 1.5

- (ii) Count the number of cells in the $1\text{ mm} \times 1\text{ mm}$ section of the counting chamber shown in Fig. 1.5.

Use your answer to calculate the number of cells per cm^3 of the suspension.

Show all your working.

number of cells per cm^3 of suspension [3]

- (iii) The scientists decided that using the Secchi stick was a less accurate method for determining the number of cells per cm^3 of suspension.

Give **two** reasons why using the Secchi stick is less accurate than using a counting chamber.

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[Total: 20]